

Week 3 – NumPy and Pandas for Data Manipulation

Objective

The objective of Week 3 was to understand the use of NumPy and Pandas for efficient data manipulation and analysis. These libraries are widely used in data science for numerical computations and handling structured datasets.

1. NumPy Arrays

NumPy provides powerful array objects that allow efficient numerical computations. NumPy arrays are faster and more efficient than traditional Python lists for mathematical operations.

Example:

```
import numpy as np
arr = np.array([10, 20, 30, 40])
```

2. NumPy Operations

NumPy allows mathematical operations on entire arrays.

Example:

```
numbers = np.array([10, 20, 30])
average = np.mean(numbers)
```

Broadcasting allows arithmetic operations between arrays and scalars.

Example:

```
numbers * 2
```

3. Pandas Data Structures

Series

A Pandas Series is a one-dimensional labeled array capable of holding different types of data.

Example:

```
import pandas as pd
s = pd.Series([1, 2, 3])
```

DataFrame

A DataFrame is a two-dimensional data structure similar to a table with rows and columns.

Example:

```
data = {"A": [1, 2], "B": [3, 4]}
df = pd.DataFrame(data)
```

4. Data Manipulation with Pandas

Pandas provides functions to clean and transform datasets.

Common operations include:

- Removing missing values (dropna())
- Removing duplicates (drop_duplicates())
- Calculating averages (mean())
- Indexing and selecting data

These operations help prepare data for analysis.

Program Implemented – Data Manipulation

Description

A Python script was developed using NumPy and Pandas to demonstrate data manipulation techniques. The dataset was cleaned by removing duplicate rows and missing values. After cleaning the dataset, the average score was calculated to demonstrate data aggregation.

Program Code

```
import numpy as np
import pandas as pd
# ----- NumPy Operations -----
print("NumPy Operations")
numbers = np.array([10, 20, 30, 40, 50])
print("Original Array:", numbers)
# Calculate average
average = np.mean(numbers)
print("Average:", average)
# Broadcasting example
result = numbers * 2
print("Array after broadcasting (*2):", result)
# ----- Pandas Data Manipulation -----
print("\nPandas DataFrame Operations")
data = {
    "Name": ["Asha", "Ravi", "Kiran", "Meena", "Ravi"],
```

```

    "Age": [21, 22, 23, None, 22],
    "Score": [85, 90, 88, 92, 90]
}
df = pd.DataFrame(data)
print("\nOriginal Dataset:")
print(df)
# Remove duplicate rows
df = df.drop_duplicates()
# Remove missing values
df = df.dropna()
print("\nCleaned Dataset:")
print(df)
# Calculate average score
average_score = df["Score"].mean()
print("\nAverage Score:", average_score)

```

OUTPUT:

```

PS C:\Users\HP\Documents\Data Science Month-1 ( D. Pravallika)\Week - 3> python Data_manipulation.py
NumPy Operations
Original Array: [10 20 30 40 50]
Average: 30.0
Array after broadcasting (*2): [ 20  40  60  80 100]

Pandas DataFrame Operations

Original Dataset:
   Name  Age  Score
0  Asha  21.0    85
1  Ravi  22.0    90
2  Kiran  23.0    88
3  Meena   NaN    92
4  Ravi  22.0    90

Cleaned Dataset:
   Name  Age  Score
0  Asha  21.0    85
1  Ravi  22.0    90
2  Kiran  23.0    88

Average Score: 87.66666666666667

```

Conclusion

Week 3 provided a strong understanding of NumPy and Pandas libraries. It demonstrated how to efficiently manipulate, clean, and analyze structured datasets, which are essential skills in data science.